

Physics Practice: RC Circuit Derivations

Grade 12 — Capacitor Charging Phase

Technical Context: Circuit Parameters

A series circuit is constructed to charge a capacitor. The circuit contains:

- An ideal DC voltage source delivering a constant electromotive force (emf) E .
- An ohmic conductor (resistor) of resistance R .
- A capacitor of capacitance C , initially uncharged ($q = 0$ at $t = 0$).
- A switch S , which is closed at time $t = 0$.

Let i be the charging current, u_C the voltage across the capacitor, and u_R the voltage across the resistor.

Task: Based on the Law of Addition of Voltages (Kirchhoff's Loop Rule), $E = u_R + u_C$, derive the following differential equations. Show all mathematical steps clearly.

1. a) Differential equation in terms of the capacitor voltage, u_C :
b) Deduce the instantaneous capacitor charge:
2. Differential equation in terms of the capacitor charge, q :
3. a) Differential equation in terms of the charging current, i :
b) Deduce the instantaneous voltage across the resistor:
4. Differential equation in terms of the resistor voltage, u_R :

